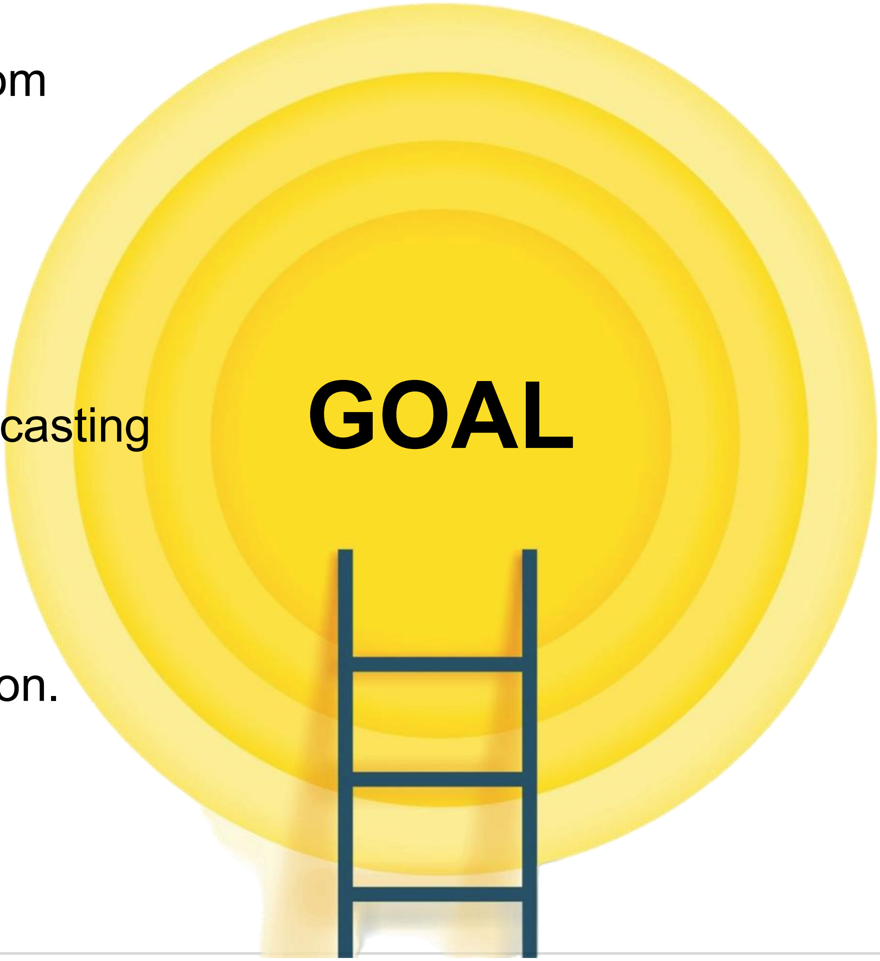
Renewable Energy Forecasting with Machine Learning

Learning Objectives

- Use machine learning to forecast solar or wind energy generation from weather data.
- Build a complete ML workflow: data collection, cleaning, feature engineering, model training, and testing.
- Compare models using RMSE, MAE, and R^2 to choose the best forecasting method.
- Identify key weather factors that impact renewable energy output.
- Visualize predicted vs. actual energy generation for easy interpretation.
- Provide clean, reproducible code and documentation for future improvements.



Source : www.freepik.com/

Tools and Technology used

1. **Python & Jupyter Notebook** for data analysis and model development.
2. **ML Libraries:** scikit-learn, Pandas, NumPy for forecasting and algorithms (e.g:regression, tree models).
3. **Visualization Tools:** Matplotlib/Seaborn for graphs and predictions.
4. **Data Sources:** NASA POWER, NREL, or CSV-based weather & energy datasets.
5. **Version Control:** Git and GitHub (GitHub repo ”***soham10667/Renewable-energy-prediction”***) to manage code and collaborate.

Methodology

1. **Data Gathering:** Collect weather and renewable energy generation data (e.g., temperature, solar irradiance, wind speed, humidity) from NASA POWER, NREL, or CSV datasets.
2. **Data Cleaning & Preparation:** Handle missing values, remove noise, create time-series features, and scale numerical variables for model readiness.
3. **Model Training:** Train multiple forecasting models (Random Forest, Gradient Boosting, LSTM) to predict solar/wind energy output from weather inputs.
4. **Evaluation:** Compare model performance using metrics like RMSE, MAE, and R^2 to select the most accurate and stable model.
5. **Deployment:** Build a simple web dashboard (e.g., Streamlit/Flask) to visualize predictions and allow users to test different weather scenarios.

Problem Statement:

1. **Renewable energy variability:** Solar and wind power are highly weather-dependent, causing unpredictable energy output. Accurate forecasting is essential for stable grid management.
2. **Grid stability challenges:** Inconsistent renewable generation creates difficulties in balancing supply and demand. Reliable predictions help optimize energy storage, scheduling, and distribution.
3. **Industry demand:** Energy companies and smart-grid operators require data-driven forecasting tools to integrate more renewable sources while improving efficiency and reducing reliance on fossil backup systems.

Solution:

1. **Renewable-Energy-ML Model:** A supervised time-series forecasting model predicts solar or wind energy output based on weather inputs such as temperature, wind speed, humidity, and solar irradiance.
2. **Actionable Insights:** The system identifies which weather factors most influence energy generation, helping grid operators optimize energy usage, storage, and scheduling.
3. **User Interface:** An interactive dashboard (e.g., Streamlit/Flask) allows non-technical users to input conditions and see predicted energy output along with visual graphs.
4. **Open Resources:** The full workflow—including data pipeline, preprocessing, trained models, and code—is documented so others can review, reuse, and build upon the project.

Screenshot of Output:

The user first supplies these meteorological parameters, after which the system runs a prediction algorithm (for example, regression or a trained ML model). The “Prediction Result” section conceptually represents the model’s quantitative output, while the “Prediction Trend” chart is intended to display changes in predicted power over multiple evaluations or time.

The layout demonstrates the theoretical pipeline of data-driven energy forecasting: raw weather data → feature inputs → model inference → visualized power predictions for decision support.

The predicted power output in kilowatts is shown numerically to summarize the model’s single best estimate for the given conditions.

The line chart visualizes how predicted output evolves over multiple runs or time steps, illustrating trends and sensitivity to changing inputs.

Overall, the dashboard demonstrates the concept of data-driven energy forecasting, where meteorological data are transformed into actionable power predictions for planning and operations.

Conclusion:

• **Impact:** This project shows how AI can improve renewable energy forecasting, helping utilities optimize power generation, reduce reliance on fossil backups, and support a cleaner, more stable energy grid—aligning with global goals of expanding sustainable energy use.

• **Educational Value:** The project reflects Edunet's mission of encouraging *youth innovation*, demonstrating how students can apply data science and machine learning to tackle real-world green-energy challenges.

• **Next Steps:** Users can explore the project's GitHub repository for datasets, code, model files, and documentation. The project is fully open-source, encouraging collaboration, experimentation, and further improvements in renewable energy prediction.